

Kimera remembers what conventional AI memory forgets.

Standard AI memory — including retrieval-augmented generation (RAG) — treats the past as an unordered **bag**: it loses order, loses history, and loses anything beyond its context window. Kimera's memory is a **landscape** that keeps the marks of everything it has been through. Each claim below was pre-registered, run on real data, and sealed as a signed proof anyone can re-verify.

Substrate: **kimera-swm** Measured by Ophamin (empirical observatory) Date: 2026-05-22
Every claim independently verifiable

WHAT IS PROVEN — ON REAL DATA

1 · It recalls beyond the context window

VALIDATED

In a long single-pass stream, Kimera still recalls experiences from far earlier — past a fixed 256-item window, the boundary where fixed-window systems go blind.

RECALL BEYOND WINDOW

1.00

PRE-REGISTERED BAR

≥ 0.50

verify: ophamin proof verify proofs/scientific/memory-horizon-flow/2026-05-22_validated_ef1cabdc
a834/proof.json · id ef1cabdca834

2 · Its memory accumulates and never resets

VALIDATED

Every time a known input returns, it lands on a measurably deeper, more-scarred place on the manifold. History is retained and compounded, not overwritten.

DEEPER EVERY TIME

1.00

PRE-REGISTERED BAR

≥ 1.00

verify: ophamin proof verify proofs/scientific/memory-permanence-flow/2026-05-22_validated_d5a4c
e3a7c13/proof.json · id d5a4ce3a7c13

3 · Order leaves a permanent mark — where RAG is blind

VALIDATED

The same documents in a different order produce a different memory address in Kimera. A standard set-based retriever returns the identical result either way — it cannot tell the orderings apart.

KIMERA (ORDER SHIFT)

0.93

RAG BASELINE

0.00

verify: ophamin proof verify proofs/scientific/memory-order-hysteresis/2026-05-22_validated_f6af1a5c15df/proof.json · id f6af1a5c15df

4 • It distinguishes risk that RAG cannot see (finance)

VALIDATED

Two orderings of the *same* set of real market returns carry genuinely different risk (maximum drawdown). Kimera separates them; a set-based retriever represents the two as identical.

KIMERA SEPARATION

1.00

RAG BASELINE

0.00

verify: ophamin proof verify proofs/scientific/finance-history-discrimination/2026-05-22_validated_7f6a839eb599/proof.json · id 7f6a839eb599

WHAT IS NOT PROVEN (STATED UP FRONT)

- A partial cue of a *seen* item did not yet recall its content better than an unseen item (memory-lift -0.08) — **inconclusive**.
- Tracking order-dependent maximum drawdown by correlation is weak (Spearman 0.18) — **inconclusive**.

THE WHOLE RECORD (FAILURES KEPT, NOT HIDDEN)

65

signed proofs

48

validated

12

refuted

5

inconclusive

WHY THESE PROOFS ARE HARD TO WAVE AWAY

Pre-registered

Signed

Real data

Reproducible

Claim, threshold and analysis plan are locked and timestamped *before* the run — no moving the goalposts.

Each result is HMAC-SHA256 sealed; altering one byte breaks the seal.

Run against real corpora and real market returns, not toy inputs.

Provenance-tracked (W3C PROV-O); re-verify any claim with one command.

Verify any claim yourself: install Ophamin, then run the `ophamin proof verify <path>` command shown under it. A valid signature confirms the result has not been altered since it was sealed. This brief was generated from the live proof store — no figure here was typed by hand.